

Optimization of MIMO STEEP for Secure Communications over MIMOME Channels

Dinglin Yu and Yingbo Hua
Department of Electrical and Computer Engineering
University of California at Riverside
Riverside, CA 92521, USA
dinglin.yu@email.ucr.edu and yhua@ucr.edu

Abstract—A transmission scheme for secure communications, called secret-message transmission by echoing encrypted probes (STEPP), can deliver a positive secrecy rate, even when eavesdropping channels are much stronger than that between users, subject to sufficient asymmetric power allocations from users. This paper considers an optimization problem for STEEP applied to MIMO (or MIMOME) channels between multi-antenna users and multi-antenna Eve. We reveal for the first time a significant increase of the achievable secrecy rate of MIMO STEEP using optimized precoders.

Index Terms—Secure communications, physical-layer security, MIMO precoder design.

I. INTRODUCTION

Secure communication without a pre-existing secure channel or a pre-shared secret key has been a long-standing problem, which has been handled by two primary approaches. The first is based on some discrete hard-to-invert function for two (or more) parties to generate their own public keys from their private keys, where the public keys are then exchanged to generate a shared secret key to be used for subsequent secure communications. Any infrastructure or standardization based on this approach is however vulnerable to future attacks by increasingly powerful computers. The other approach, also known as physical layer security (PLS), is based on information-theoretical security, which is not breakable by any computing power. This paper is concerned with the second approach.

There are numerous works on PLS in the literature, and they can be classified into two groups: one is based on single-trip wiretap channel (ST-WTC) schemes where a secret-message is transmitted from one user to another via a single-trip coding scheme, and the other is based on secret-key generation (SKG) where correlated data sets collected at two users are used to generate a shared secret key.

It is known [1] and [2] that, when using ST-WTC schemes, the achievable secrecy rate (in bits per channel use) from Bob to Alice is zero whenever Eve's receive channel capacity from Bob cannot be made larger than Alice's receive channel capacity from Bob. Such a situation often happens when Eve's distance to Bob is shorter than Alice's and at the same time Eve's number of antennas is no less than Bob's.

On the other hand, it is also known [3] and [4] that the secret-key capacity (in bits per probing channel use) for SKG from channel probing and information reconciliation does not have the above mentioned limitation. But secret-key capacity does not take into account the channel uses needed for information reconciliation (IR). Many proposed methods in the literature for IR require a substantial amount of additional channel uses.

An early attempt to take advantage of both SKG and WTC was shown in [5], where the authors considered binary signaling over SISO channels. Recently, a much more general fusion of SKG and WTC has resulted in what is called secret-message transmission by echoing encrypted probes (STEPP) [6], [7], [8], [9], [10], [11]. Unlike any ST-WTC scheme, STEEP utilizes a round-trip (RT) coding scheme inspired by an encryption lemma shown in [12] and [13], which allows STEEP to yield a positive secrecy rate by an asymmetric power control under virtually any channel conditions. In other words, via a RT coding scheme, STEEP creates a virtual WTC system where Eve's effective receive channel is always disadvantaged even if her original or raw receive channels are stronger than users'. The above property holds in general, including the case where STEEP is applied to MIMO channels.

However, the current MIMO version of STEEP as shown in [7] is not optimized, which does not exploit any possible knowledge of Eve's channel matrices. The aim of this paper is to show how to optimize the precoding matrices for MIMO STEEP, assuming that all channel matrices are given. This will allow us to see the secrecy rate gap between the optimized MIMO STEEP and a baseline MIMO STEEP.

It is important to note that for the MIMO ST-WTC scheme as studied in [1] and [2], there are many prior works on optimization of the transmit covariance matrix to maximize its secrecy rate. See [14], [15], [16] and the references therein. But the secrecy rate expression for the MIMO STEEP is very much different from that for the MIMO ST-WTC. The former is more complicated than the latter. What is similar here to prior works is that we also apply such principles as alternating optimization (AO), weighted minimum mean squared error (WMMSE), successive convex approximation (SCA), and projected gradient ascent (PGA).

II. CHANNEL MODEL AND MIMO STEEP

Consider a network of MIMO channels among two legitimate users (Alice and Bob) and an eavesdropper (Eve). Alice and Bob are equipped with n_1 and n_2 antennas, respectively. Eve possesses l_1 antennas for intercepting in the probing phase, and l_2 antennas for intercepting in the echoing phase. We adopt a block-fading narrowband channel model where all channel matrices are constant within each channel coherence period of T_c second while all signals and noises at different sampling/symbol times within each T_c are modelled as independent and identically distributed (i.i.d.).

Let $\mathbf{x}_1(k)$ be the k th symbol transmitted by Alice in phase 1 (probing phase), and the corresponding signals received by Bob and Eve are respectively $\mathbf{y}_2(k)$ and $\mathbf{z}_1(k)$:

$$\begin{aligned}\mathbf{y}_2(k) &= \mathbf{H}_2 \mathbf{x}_1(k) + \mathbf{w}_2(k), \\ \mathbf{z}_1(k) &= \mathbf{G}_1 \mathbf{x}_1(k) + \mathbf{v}_1(k).\end{aligned}\quad (1)$$

Let $\mathbf{x}_2(k)$ be the signal transmitted by Bob in phase 2 (echoing phase), and the corresponding signals received by Alice and Eve are respectively $\mathbf{y}_1(k)$ and $\mathbf{z}_2(k)$:

$$\begin{aligned}\mathbf{y}_1(k) &= \mathbf{H}_1 \mathbf{x}_2(k) + \mathbf{w}_1(k), \\ \mathbf{z}_2(k) &= \mathbf{G}_2 \mathbf{x}_2(k) + \mathbf{v}_2(k).\end{aligned}\quad (2)$$

Here $\mathbf{H}_2 \in \mathbb{C}^{n_2 \times n_1}$, $\mathbf{H}_1 \in \mathbb{C}^{n_1 \times n_2}$, $\mathbf{G}_1 \in \mathbb{C}^{l_1 \times n_1}$, $\mathbf{G}_2 \in \mathbb{C}^{l_2 \times n_2}$ are the corresponding channel matrices. All noise vectors $\mathbf{w}_1(k)$, $\mathbf{w}_2(k)$, $\mathbf{v}_1(k)$ and $\mathbf{v}_2(k)$ are assumed to be $\mathcal{CN}(0, \mathbf{I})$. We do not assume full-duplex transceivers.

As a generalization of the round-trip precoding for Gaussian signaling used in STEEP shown in [7], we now let

$$\begin{aligned}\mathbf{x}_1(k) &= \mathbf{W}_1 \mathbf{s}_1(k), \\ \mathbf{x}_2(k) &= \mathbf{W}_y \mathbf{y}_2(k) + \mathbf{W}_s \mathbf{s}_2(k),\end{aligned}\quad (3)$$

where $\mathbf{s}_1(k)$ is the probing signal, $\mathbf{s}_2(k)$ is the secret-message dependent vector, and both are assumed to be $\mathcal{CN}(0, \mathbf{I})$. Here $\mathbf{W}_1 \in \mathbb{C}^{n_1 \times n_1}$, $\mathbf{W}_y \in \mathbb{C}^{n_2 \times n_2}$ and $\mathbf{W}_s \in \mathbb{C}^{n_2 \times n_2}$ are the round-trip precoding matrices.

Following a similar analysis as shown in [7], we can show that an achievable secrecy rate of the above MIMO STEEP scheme in terms of $\mathbf{s}_2(k)$ from Bob to Alice, in bits per channel use or bits/s/Hz, is $\frac{1}{2}R_s^+ = \max(0, \frac{1}{2}R_s)$ with

$$R_s = C_U - C_E \quad (4)$$

where C_U and C_E are respectively the effective capacities at Alice and Eve to receive the information in $\mathbf{s}_2(k)$, i.e.,

$$\begin{aligned}C_U &= \mathbb{I}(\mathbf{s}_2(k); \mathbf{x}_1(k), \mathbf{y}_1(k)) = \log \frac{1}{|\mathbf{K}_{\Delta s_2 | x_1, y_1}|} \\ &= \log \frac{|\mathbf{H}_1 \mathbf{W}_y \mathbf{W}_y^H \mathbf{H}_1^H + \mathbf{H}_1 \mathbf{W}_s \mathbf{W}_s^H \mathbf{H}_1^H + \mathbf{I}|}{|\mathbf{H}_1 \mathbf{W}_y \mathbf{W}_y^H \mathbf{H}_1^H + \mathbf{I}|},\end{aligned}\quad (5)$$

$$\begin{aligned}C_E &= \mathbb{I}(\mathbf{s}_2(k); \mathbf{z}_1(k), \mathbf{z}_2(k)) = \log \frac{1}{|\mathbf{K}_{\Delta s_2 | z_1, z_2}|} \\ &= \log \frac{|\mathbf{G}_2 \mathbf{W}_y \mathbf{K} \mathbf{W}_y^H \mathbf{G}_2^H + \mathbf{G}_2 \mathbf{W}_s \mathbf{W}_s^H \mathbf{G}_2^H + \mathbf{I}|}{|\mathbf{G}_2 \mathbf{W}_y \mathbf{K} \mathbf{W}_y^H \mathbf{G}_2^H + \mathbf{I}|},\end{aligned}\quad (6)$$

with $|\cdot| \doteq \det(\cdot)$ and

$$\begin{aligned}\mathbf{K} &\doteq \mathbf{K}_{\Delta y_2 | z_1} = \mathbf{H}_2 \mathbf{W}_1 \mathbf{W}_1^H \mathbf{H}_2^H + \mathbf{I} \\ &\quad - \mathbf{H}_2 \mathbf{W}_1 \mathbf{W}_1^H \mathbf{G}_1^H (\mathbf{G}_1 \mathbf{W}_1 \mathbf{W}_1^H \mathbf{G}_1^H + \mathbf{I})^{-1} \\ &\quad \cdot \mathbf{G}_1 \mathbf{W}_1 \mathbf{W}_1^H \mathbf{H}_2^H.\end{aligned}\quad (7)$$

Here $\mathbf{K}_{\Delta y_2 | z_1}$ (for example) denotes the MSE matrix of the MMSE estimate of $\mathbf{y}_2$ from $\mathbf{z}_1$.

Assuming $n_1 \geq n_2$, the precoders considered in [7] are equivalent to

$$\begin{cases} \mathbf{W}_1 = \sqrt{\frac{p_1}{n_1}} \mathbf{I}_{n_1}, \\ \mathbf{W}_y = \sqrt{a} \sqrt{\frac{p_1}{n_1} \Sigma_2 (\frac{p_1}{n_1} \Sigma_2^2 + \mathbf{I}_{n_2})^{-1} \mathbf{U}_2^H}, \\ \mathbf{W}_s = \sqrt{a} \mathbf{I}_{n_2}, \end{cases}\quad (8)$$

where Σ_2 and $\mathbf{U}_2$ are from the (thin) SVD $\mathbf{H}_2 = \mathbf{U}_2 \Sigma_2 \mathbf{V}_2^H$, and a is such that

$$a \left(\text{Tr} \left(\frac{p_1}{n_1} \Sigma_2^2 \left(\frac{p_1}{n_1} \Sigma_2^2 + \mathbf{I}_{n_2} \right)^{-1} \right) + n_2 \right) = p_2. \quad (9)$$

The above precoders depend on $\mathbf{H}_2$ but are invariant to $(\mathbf{H}_1, \mathbf{G}_1, \mathbf{G}_2)$, which will be referred as a baseline choice. It is shown in [7] that, even using the baseline precoders, R_s can be made positive by using sufficiently large p_1 and $\frac{p_2}{p_1}$ for virtually any given $(\mathbf{H}_1, \mathbf{H}_2, \mathbf{G}_1, \mathbf{G}_2)$. This property of MIMO STEEP is in great contrast to the conventional (single-trip) MIMO WTC schemes, the latter of which cannot produce a positive secrecy rate for a large channel space of $(\mathbf{H}_1, \mathbf{H}_2, \mathbf{G}_1, \mathbf{G}_2)$ if $l_1 \geq n_1$ and $l_2 \geq n_2$ [1], [2].

In this paper, we explore the performance of MIMO STEEP under optimized precoders. Namely, we are interested in the following problem for MIMO STEEP (which may also be viewed as MIMO RT-WTC problem):

$$\max_{\mathbf{W}_1, \mathbf{W}_y, \mathbf{W}_s} R_s, \quad (10a)$$

$$\text{s.t.} \quad \text{Tr}(\mathbf{W}_1 \mathbf{W}_1^H) \doteq p_1^* \leq p_1, \quad (10b)$$

$$\text{Tr}(\mathbf{W}_y \mathbf{K}_{y_2} \mathbf{W}_y^H) + \text{Tr}(\mathbf{W}_s \mathbf{W}_s^H) \doteq p_2^* \leq p_2, \quad (10c)$$

where $\mathbf{K}_{y_2} = \mathbf{H}_2 \mathbf{W}_1 \mathbf{W}_1^H \mathbf{H}_2^H + \mathbf{I}$, and p_1 and p_2 are the allocated upper bounds on powers (or “allowed” powers) at Alice and Bob respectively while p_1^* and p_2^* denote the “actual” powers consumed by Alice and Bob.

III. OPTIMIZATION

To solve the problem (10), we first notice that with fixed $\mathbf{W}_1$, the constraint (10c) is convex while (10b) is not needed. With fixed $\mathbf{W}_2 \doteq (\mathbf{W}_y, \mathbf{W}_s)$, both (10b) and (10c) are convex constraints. This makes AO a natural choice. So we propose to optimize $\mathbf{W}_1$ and $\mathbf{W}_2$ alternately.

With fixed $\mathbf{W}_1$, R_s remains non-concave of $\mathbf{W}_2$. But its structure allows the application of WMMSE and SCA as shown later. With fixed $\mathbf{W}_2$, the structure of R_s in terms of $\mathbf{W}_1$ does not allow the use of WMMSE, and in this case we will apply the more general principle PGA.

A. Optimizing $(\mathbf{W}_y, \mathbf{W}_s)$ given $\mathbf{W}_1$

With given $\mathbf{W}_1$, we can write

$$\tilde{\mathbf{H}}_2 \doteq \mathbf{H}_2 \mathbf{W}_1, \quad \tilde{\mathbf{G}}_1 \doteq \mathbf{G}_1 \mathbf{W}_1, \quad (11)$$

and $\mathbf{K}$ and $\mathbf{K}_{y_2}$ are constant matrices:

$$\begin{aligned} \mathbf{K} &= \tilde{\mathbf{H}}_2 \tilde{\mathbf{H}}_2^H + \mathbf{I} - \tilde{\mathbf{H}}_2 \tilde{\mathbf{G}}_1^H (\tilde{\mathbf{G}}_1 \tilde{\mathbf{G}}_1^H + \mathbf{I})^{-1} \tilde{\mathbf{G}}_1 \tilde{\mathbf{H}}_2^H, \\ \mathbf{K}_{y_2} &= \tilde{\mathbf{H}}_2 \tilde{\mathbf{H}}_2^H + \mathbf{I}. \end{aligned} \quad (12)$$

In this case, (10) reduces to

$$\max_{(\mathbf{W}_y, \mathbf{W}_s) \in \mathcal{C}_2} R_s, \quad \mathcal{C}_2 = \{(\mathbf{W}_y, \mathbf{W}_s) : (10c)\}. \quad (13)$$

The objective function of the problem (13) can be written as:

$$R_s = \underbrace{[\log |\mathbf{J}_1| + \log |\mathbf{J}_4|]}_{T_{good}} - \underbrace{[\log |\mathbf{J}_2| + \log |\mathbf{J}_3|]}_{T_{bad}}, \quad (14)$$

where

$$\begin{aligned} \mathbf{J}_1 &= \mathbf{H}_1 \mathbf{W}_y \mathbf{W}_y^H \mathbf{H}_1^H + \mathbf{H}_1 \mathbf{W}_s \mathbf{W}_s^H \mathbf{H}_1^H + \mathbf{I}, \\ \mathbf{J}_2 &= \mathbf{H}_1 \mathbf{W}_y \mathbf{W}_y^H \mathbf{H}_1^H + \mathbf{I}, \\ \mathbf{J}_3 &= \mathbf{G}_2 \mathbf{W}_y \mathbf{K} \mathbf{W}_y^H \mathbf{G}_2^H + \mathbf{G}_2 \mathbf{W}_s \mathbf{W}_s^H \mathbf{G}_2^H + \mathbf{I}, \\ \mathbf{J}_4 &= \mathbf{G}_2 \mathbf{W}_y \mathbf{K} \mathbf{W}_y^H \mathbf{G}_2^H + \mathbf{I}. \end{aligned} \quad (15)$$

Here $\log |\mathbf{J}|$ is a concave function of $\mathbf{J}$. But R_s is a not concave function of $\mathbf{W}_y$ and $\mathbf{W}_s$. And the problem (13) is still non-convex. To tackle this, we next apply the principles of SCA and WMMSE.

1) *Applying SCA:* We now apply SCA to the last two terms in (14). Since $-\log |\mathbf{J}|$ is a convex function of $\mathbf{J}$, it is lower bounded by its “surrogate” as follows:

$$\begin{aligned} -\log |\mathbf{J}| &\geq -\log |\mathbf{J}^{(k)}| \\ &\quad - \text{Tr} \left((\nabla_{\mathbf{J}} \log |\mathbf{J}^{(k)}|)^H (\mathbf{J} - \mathbf{J}^{(k)}) \right) \\ &= -\log |\mathbf{J}^{(k)}| - \text{Tr} (\mathbf{J}^{(k)-1} (\mathbf{J} - \mathbf{J}^{(k)})) \\ &= \underbrace{-\log |\mathbf{J}^{(k)}| + \text{Tr}(\mathbf{I})}_{\text{const}} - \text{Tr} (\mathbf{J}^{(k)-1} \mathbf{J}) \doteq T_{surro}^{(k)}, \end{aligned} \quad (16)$$

where $\mathbf{J}^{(k)}$ denotes the value of the matrix $\mathbf{J}$ evaluated at the k th iteration of $\mathbf{W}_y$ and $\mathbf{W}_s$, and we have used $\nabla_{\mathbf{X}} \log |\mathbf{X}| = \mathbf{X}^{-1}$ [17].

Applying (16) to $-\log |\mathbf{J}_3|$, we have

$$\begin{aligned} T_3^{surro} &= \text{const} - \text{Tr} (\mathbf{W}_y^H \mathbf{G}_2^H \mathbf{J}_3^{(k)-1} \mathbf{G}_2 \mathbf{W}_y \mathbf{K}) \\ &\quad - \text{Tr} (\mathbf{W}_s^H \mathbf{G}_2^H \mathbf{J}_3^{(k)-1} \mathbf{G}_2 \mathbf{W}_s). \end{aligned} \quad (17)$$

Similarly, for $-\log |\mathbf{J}_2|$, we have

$$T_2^{surro} = \text{const} - \text{Tr} (\mathbf{W}_y^H \mathbf{H}_1^H \mathbf{J}_2^{(k)-1} \mathbf{H}_1 \mathbf{W}_y). \quad (18)$$

Therefore, with the k th iteration of $\mathbf{W}_y$ and $\mathbf{W}_s$, the last two terms (or T_{bad}) in (14) can be replaced by

$$\begin{aligned} T^{surro} &= -\text{Tr} (\mathbf{W}_y^H \mathbf{G}_2^H \mathbf{J}_3^{(k)-1} \mathbf{G}_2 \mathbf{W}_y \mathbf{K}) \\ &\quad - \text{Tr} (\mathbf{W}_s^H \mathbf{G}_2^H \mathbf{J}_3^{(k)-1} \mathbf{G}_2 \mathbf{W}_s) \\ &\quad - \text{Tr} (\mathbf{W}_y^H \mathbf{H}_1^H \mathbf{J}_2^{(k)-1} \mathbf{H}_1 \mathbf{W}_y), \end{aligned} \quad (19)$$

where all constant terms have been dropped.

2) *Applying WMMSE:* We now apply the principle of WMMSE to the first two terms in (14). For $\mathbf{J}_1$, we define

$$\mathbf{V}_1 \doteq [\mathbf{W}_y \quad \mathbf{W}_s] \in \mathbb{C}^{n_2 \times (2n_2)}. \quad (20)$$

Then

$$\mathbf{J}_1 = \mathbf{I} + \mathbf{H}_1 \mathbf{V}_1 \mathbf{V}_1^H \mathbf{H}_1^H. \quad (21)$$

For $\mathbf{J}_4$, we let $\mathbf{K} = \mathbf{L} \mathbf{L}^H$ and define

$$\mathbf{V}_4 \doteq \mathbf{W}_y \mathbf{L} \in \mathbb{C}^{n_2 \times n_2}, \quad (22)$$

so that

$$\mathbf{J}_4 = \mathbf{I} + \mathbf{G}_2 \mathbf{V}_4 \mathbf{V}_4^H \mathbf{G}_2^H. \quad (23)$$

Define $\mathbf{y} = \mathbf{H} \mathbf{V} \mathbf{s} + \mathbf{n}$ with $\mathbf{s} \sim \mathcal{CN}(\mathbf{0}, \mathbf{I})$ and $\mathbf{n} \sim \mathcal{CN}(\mathbf{0}, \mathbf{I})$. A linear receiver $\hat{\mathbf{s}} = \mathbf{U}^H \mathbf{y}$ has the following MSE matrix

$$\begin{aligned} \mathbf{E}(\mathbf{U}, \mathbf{V}) &\doteq \mathbb{E}[(\mathbf{s} - \mathbf{U}^H \mathbf{y})(\mathbf{s} - \mathbf{U}^H \mathbf{y})^H] \\ &= \mathbf{I} - \mathbf{U}^H \mathbf{H} \mathbf{V} - (\mathbf{U}^H \mathbf{H} \mathbf{V})^H + \mathbf{U}^H (\mathbf{H} \mathbf{V} \mathbf{V}^H \mathbf{H}^H + \mathbf{I}) \mathbf{U}. \end{aligned} \quad (24)$$

Recall the WMMSE identity [15], [18]:

$$\begin{aligned} &\log |\mathbf{I} + \mathbf{H} \mathbf{V} \mathbf{V}^H \mathbf{H}^H| \\ &= \max_{\mathbf{U}, \mathbf{S} \succ \mathbf{0}} \log |\mathbf{S}| - \text{Tr}(\mathbf{S} \mathbf{E}(\mathbf{U}, \mathbf{V})) + d, \end{aligned} \quad (25)$$

where d is the number of columns of $\mathbf{V}$. Given $\mathbf{S}$ and $\mathbf{U}$, $-\text{Tr}(\mathbf{S} \mathbf{E}(\mathbf{U}, \mathbf{V}))$ is a concave quadratic function of $\mathbf{V}$.

Applying (25) to $\log |\mathbf{J}_1|$ and $\log |\mathbf{J}_4|$ yields

$$\begin{aligned} \log |\mathbf{J}_1| &= \max_{\mathbf{U}_1, \mathbf{S}_1 \succ \mathbf{0}} \log |\mathbf{S}_1| - \text{Tr}(\mathbf{S}_1 \mathbf{E}_1(\mathbf{U}_1, \mathbf{V}_1)) + 2n_2, \\ \log |\mathbf{J}_4| &= \max_{\mathbf{U}_4, \mathbf{S}_4 \succ \mathbf{0}} \log |\mathbf{S}_4| - \text{Tr}(\mathbf{S}_4 \mathbf{E}_4(\mathbf{U}_4, \mathbf{V}_4)) + n_2, \end{aligned} \quad (26)$$

where $\mathbf{E}_1(\mathbf{U}_1, \mathbf{V}_1)$ and $\mathbf{E}_4(\mathbf{U}_4, \mathbf{V}_4)$ follow (24).

Given $(\mathbf{W}_y^{(k)}, \mathbf{W}_s^{(k)})$, we have $\mathbf{V}_1^{(k)} = [\mathbf{W}_y^{(k)} \quad \mathbf{W}_s^{(k)}]$ and $\mathbf{V}_4^{(k)} = \mathbf{W}_y^{(k)} \mathbf{L}$. And the corresponding optimal $\mathbf{U}_1$ and $\mathbf{U}_4$ are

$$\begin{aligned} \mathbf{U}_1^{(k+1)} &= (\mathbf{H}_1 \mathbf{V}_1^{(k)} (\mathbf{V}_1^{(k)})^H \mathbf{H}_1^H + \mathbf{I})^{-1} \mathbf{H}_1 \mathbf{V}_1^{(k)}, \\ \mathbf{U}_4^{(k+1)} &= (\mathbf{G}_2 \mathbf{V}_4^{(k)} (\mathbf{V}_4^{(k)})^H \mathbf{G}_2^H + \mathbf{I})^{-1} \mathbf{G}_2 \mathbf{V}_4^{(k)}. \end{aligned} \quad (27)$$

Let $\mathbf{E}_1^{(k+1)} \doteq \mathbf{E}_1(\mathbf{U}_1^{(k+1)}, \mathbf{V}_1^{(k)})$ and $\mathbf{E}_4^{(k+1)} \doteq \mathbf{E}_4(\mathbf{U}_4^{(k+1)}, \mathbf{V}_4^{(k)})$. Then the optimal weights are

$$\begin{aligned} \mathbf{S}_1^{(k+1)} &= (\mathbf{E}_1^{(k+1)})^{-1}, \\ \mathbf{S}_4^{(k+1)} &= (\mathbf{E}_4^{(k+1)})^{-1}. \end{aligned} \quad (28)$$

3) *Combining SCA and WMMSE:* Combining the WMMSE representation for T_{good} and the SCA lower bound for T_{bad} in (14), R_s with the k th update of $(\mathbf{W}_y, \mathbf{W}_s)$ can be represented by

$$\begin{aligned} g^{(k)} &= -\text{Tr} (\mathbf{S}_1^{(k+1)} \mathbf{E}_1(\mathbf{U}_1^{(k+1)}, \mathbf{V}_1)) \\ &\quad - \text{Tr} (\mathbf{S}_4^{(k+1)} \mathbf{E}_4(\mathbf{U}_4^{(k+1)}, \mathbf{V}_4)) + T_{surro}(\mathbf{W}_y, \mathbf{W}_s) \end{aligned} \quad (29)$$

which is a concave quadratic function of $(\mathbf{W}_y, \mathbf{W}_s)$. Here $\mathbf{V}_1 = [\mathbf{W}_y \ \mathbf{W}_s]$ and $\mathbf{V}_4 = \mathbf{W}_y \mathbf{L}$. Therefore, the problem (13) can be solved iteratively by solving the following convex problem:

$$\max_{\mathbf{W}_y, \mathbf{W}_s \in \mathcal{C}_2} g^{(k)}, \quad (30)$$

which maximizes a concave quadratic objective over a convex quadratic constraint set.

B. Optimizing $\mathbf{W}_1$ given $(\mathbf{W}_y, \mathbf{W}_s)$

With given $(\mathbf{W}_y, \mathbf{W}_s)$, (10) reduces to

$$\max_{\mathbf{W}_1 \in \mathcal{C}_1} R_s, \quad \mathcal{C}_1 \doteq \{\mathbf{W}_1 : (10b), (10c)\}. \quad (31)$$

Here $\mathcal{C}_1$ is convex but the dependency of R_s on $\mathbf{W}_1$ (see (7)) does not allow the use of WMMSE. But we will use the PGA principle [20] as follows:

$$\begin{aligned} \mathbf{W}_{1,\text{tmp}} &\leftarrow \mathbf{W}_1 + \tau \nabla_{\mathbf{W}_1} R_s(\mathbf{W}_1, \mathbf{W}_y, \mathbf{W}_s) \\ \mathbf{W}_1 &\leftarrow \Pi_{\mathcal{C}_1}(\mathbf{W}_{1,\text{tmp}}) \end{aligned} \quad (32)$$

where τ is chosen by Armijo backtracking [19], and $\Pi_{\mathcal{C}_1}(\cdot)$ denotes the projection onto $\mathcal{C}_1$.

By alternate optimization (AO) between $(\mathbf{W}_y, \mathbf{W}_s)$ and $\mathbf{W}_1$, we can obtain locally optimal $(\mathbf{W}_1, \mathbf{W}_y, \mathbf{W}_s)$.

IV. SIMULATION

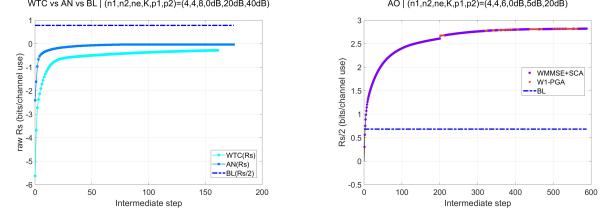
To illustrate the results of the optimized secrecy rate of MIMO STEEP, we consider a scenario where Eve is at the middle between Alice and Bob. The distance between Alice and Bob is normalized to be one, and the channel matrices between them are subject to Rician fading. Specifically, we assume

$$\mathbf{H}_2 = \sqrt{K/(K+1)} \mathbf{a}_r \mathbf{a}_t^H e^{j\theta_2} + \sqrt{1/(K+1)} \mathbf{N}_2 \quad (33)$$

where K is the Rician factor, $\mathbf{N}_2$ consists of i.i.d. elements from $\mathcal{CN}(0, 1)$, θ_2 is a uniform random phase, and $\mathbf{a}_r$ and $\mathbf{a}_t$ are receive and transmit steering vectors of linear arrays. The choice of $\mathbf{H}_1$ is similar. But $\mathbf{G}_1$ and $\mathbf{G}_2$ are scaled up by the factor 2 (assuming pathloss exponent equal to 2) while they have the similar structures as $\mathbf{H}_1$ and $\mathbf{H}_2$. Also assume that all arrays are oriented toward each other (so that the $\mathbf{a}$ -vectors consist of all ones). Also let $n_E = l_1 = l_2$, $K = 1$ (or 0dB), and recall $\mathbf{W}_2 \doteq (\mathbf{W}_y, \mathbf{W}_s)$.

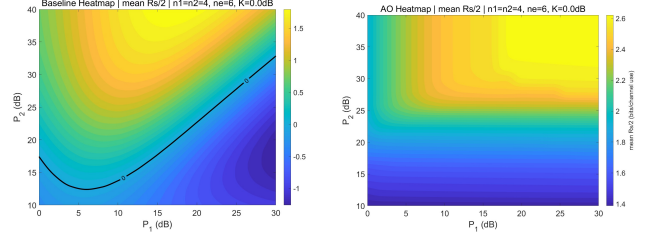
Fig. 1a shows a case where the optimized R_s of the classic ST-WTC scheme [1]-[2] (equivalently using $p_1 = 0$ here) can only converge to zero while STEEP using the baseline (BL) precoders shown in (8) and $(p_1, p_2) = (20, 40)$ dB yields a significant $\frac{1}{2} R_s > 0$. In this figure, WTC denotes the case for $\mathbf{W}_y = 0$ and optimized $\mathbf{W}_s$, and AN (artificial noise) denotes the case for optimized $(\mathbf{W}_y, \mathbf{W}_s)$. Both WTC and AN converge to the same secrecy rate (zero here). It is known that WTC and AN yield the same R_s after optimization. All three curves in Fig. 1a are based on a common random realization of $(\mathbf{H}_1, \mathbf{H}_2, \mathbf{G}_1, \mathbf{G}_2)$.

Fig. 1b shows that $\frac{1}{2} R_s$ of STEEP using the optimized precoders is much higher than that using the baseline



(a) WTC vs AN vs BL. $n_E = 8$, (b) Opt. STEEP vs BL. $n_E = 6$, $(p_1, p_2) = (20, 40)$ dB. $(p_1, p_2) = (5, 20)$ dB.

Fig. 1: Convergence patterns of R_s using $n_1 = n_2 = 4$.



(a) $\frac{1}{2} R_s$ of Baseline STEEP (b) $\frac{1}{2} R_s$ of Optimized STEEP

Fig. 2: Averaged heatmaps of $\frac{1}{2} R_s$ using $(n_1, n_2, n_E) = (4, 4, 6)$. For (a), the axes are the “actual” powers (p_1^*, p_2^*) . For (b), the axes are the “allowed” powers (p_1, p_2) .

(BL) precoders in (8), where $(n_1, n_2, n_E) = (4, 4, 6)$ and $(p_1, p_2) = (5, 20)$ dB. This figure also shows the convergence behavior of R_s when optimizing $\mathbf{W}_1$ and $\mathbf{W}_2$ alternately. The purple dots correspond to $\mathbf{W}_2$ being optimized while the red dots correspond to $\mathbf{W}_1$ being optimized.

Fig. 2a shows an averaged heatmap of $\frac{1}{2} R_s$ of the baseline STEEP versus (p_1^*, p_2^*) . For each (p_1^*, p_2^*) , 10 realizations of $(\mathbf{H}_1, \mathbf{H}_2, \mathbf{G}_1, \mathbf{G}_2)$ with $(n_1, n_2, n_E) = (4, 4, 6)$ were used. As expected, $R_s > 0$ if p_1^* and p_2^* are large enough.

Fig. 2b shows a similar heatmap of the optimized STEEP versus (p_1, p_2) . We see that R_s saturates when p_1 and p_2 are beyond some values. Since R_s typically increases with p_2^* , the optimal p_2^* is p_2 . But similar to Fig. 2a, the optimal value of p_1^* increases with p_2^* (due to “strong” Eve) but is much (about 10-15dB) smaller than p_2^* . Fig. 2b unfortunately conveys little information about the corresponding values of (p_1^*, p_2^*) , which cannot be discussed further due to page limit.

V. CONCLUSION

For the first time, we have shown how to compute the optimal precoders for MIMO STEEP, which is a round-trip scheme important for delivering a positive secrecy rate even when Eve’s channels are much stronger than users’. Among the important observations, we have found that, with the exact knowledge of Eve’s channels, the secrecy rate of MIMO STEEP using optimized precoders can be much higher than that using the baseline precoders. Future work needs to consider optimal precoder design with only limited knowledge of Eve such as her worst possible location and maximal possible number of antennas in a given application.

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